NATIONAL TEACHERS COLLEGE • SCHOOL OF ARTS, SCIENCES AND TECHNOLOGY

Capstone Documentation Format Conversion Guide

A Comprehensive Guide to Reorganizing BSIT Capstone Documentation from the Legacy 2-Chapter Layout into the Updated 3-Chapter Standard Architecture

Target Audience: BSIT Capstone Proponents & Advisers | **Source Reference:** Capstone Documentation Template - New Format (2026)

1. Executive Summary & Conversion Rationale

The School of Arts, Sciences and Technology at National Teachers College (NTC) has updated the standard Capstone Documentation Template for the Bachelor of Science in Information Technology (BSIT) program. The legacy documentation format consolidated all introductory, literature, technical, and conceptual framework sections into a single massive **Chapter I (Introduction)** and placed all system design, architecture, SDLC, requirements, and statistical treatments into **Chapter II (Methodology)**.

The updated curriculum standard establishes a modular, industry-aligned **3-Chapter structure** for Capstone Project 1 (Proposal Stage), redistributing content logically across specific domains:

- **Chapter 1: Introduction (The "Why")** — Focuses strictly on project context, purpose, objectives, scope/delimitations, significance, and operational terms.
- **Chapter 2: Review of Related Literature and Studies (The "Foundation")** — Isolates thematic literature, empirical studies, and introduces a mandatory *Synthesis* section.
- **Chapter 3: Technical Background (The "Tech Stack & Architecture")** — Dedicated to technology evaluations, existing system analysis, and detailed hardware, software, and network feasibility.
- **Chapter 4: Methodology, Testing, and Evaluation (The "How")** — Receives the SDLC, system diagrams (DFD, ERD, Use Case, Data Dictionary), testing plans, and evaluation methods previously located in Chapter II.

2. Structural Mapping Matrix (Previous vs. New Format)

The table below provides a complete side-by-side mapping between the Previous Format (as demonstrated in projects like *AgriDirect*) and the New Capstone Template Standard.

Previous Format Section	New Standard Destination	Action Type	Migration Guidelines
Ch I: Project Context & Background	**Ch 1.1 Project Context**	RELOCATE	Keep core organizational pain points, current baseline situations, and client context.
Ch I: Introduction (Problem/ Solution)	**Ch 1.2 Purpose and Description**	SPLIT / NEW SECTION	Extract general background paragraphs into an explicit problem statement and system solution summary.
Ch I: Objectives of the Study	**Ch 1.3 Objectives of the Study** • 1.3.1 General Objective • 1.3.2 Specific Objectives	RELOCATE	Ensure strict sub-numbering (1.3.1 & 1.3.2). Specific objectives must follow the "design, develop, evaluate" progression.

Previous Format Section	New Standard Destination	Action Type	Migration Guidelines
Ch I: Scope and Delimitation	**Ch 1.4 Scope and Delimitations**	RELOCATE	Maintain clear delineation between functional/geographic scope and explicit system boundaries.
Ch I: Significance of the Study	**Ch 1.5 Significance of the Study**	RELOCATE	Retain stakeholder breakdown (Beneficiaries, Client Agency, Future Researchers, Academic Institution).
Ch I: Definition of Terms	**Ch 1.6 Definition of Terms**	RELOCATE UP	Move from the very end of Ch I up to Section 1.6. Keep alphabetical ordering and operational definitions.
Ch I: Review of Related Literature	**Ch 2.1 Related Literature**	MOVE TO CH 2	Extract concepts, frameworks, and laws from Ch I. Group logically into Foreign and Local sections.
Ch I: Review of Related Studies	**Ch 2.2 Related Studies**	MOVE TO CH 2	Extract existing systems, software, and empirical research from Ch I. Group into Foreign and Local.
Not present in Previous Format	**Ch 2.3 Synthesis of Reviewed Literature & Studies**	CREATE NEW	Draft a new section identifying research gaps, contrasting existing work with your project, and highlighting key takeaways.
Ch I: Technical Background	**Ch 3.1 Overview of Current Technologies**	MOVE TO CH 3	Extract high-level technical stack descriptions (Django, MySQL, Python ML, PWA, Chart.js) into 3.1.
Implied in Ch I Context / RRL	**Ch 3.2 Evaluation of the Existing System**	CREATE NEW	Formalize analysis of manual processes or existing legacy systems, specifying bottlenecks and inefficiencies.
Ch I: IPO Inputs & Ch II: Requirements Analysis	**Ch 3.3 Technical Feasibility** • 3.3.1 Hardware Specs • 3.3.2 Software Specs	CONSOLIDATE & MOVE	Extract hardware and software specifications tables from Chapter II and IPO Model into structured sub-sections in Chapter 3.
Ch II: Software Architecture	**Ch 3.3.3 Network/Architecture Specs**	MOVE TO CH 3	Move system layer diagrams, 3-tier architecture details, and REST API/PWA network topologies into 3.3.3.
Ch II: SDLC, Diagrams, Requirements, Testing	**Chapter 4: Methodology, Testing, and Evaluation**	REASSIGN TO CH 4	Move Agile SDLC (4.1), Requirements Analysis (4.2), System Diagrams (4.3), Testing Plan (4.4), and Evaluation (4.5) to Chapter 4.

Key Organizational Takeaway

In the Previous Format, **Chapter I was bloated** with 10 disparate topics ranging from introduction to technical stack, literature, conceptual framework, and terms. The New Format creates clean separation: **Chapter 1 = Problem & Scope**, **Chapter 2 = Literature & Gaps**, **Chapter 3 = Technology & Feasibility**, and **Chapter 4 = System Design & Implementation Plan**.

3. Detailed Chapter-by-Chapter Migration Instructions

1 Converting to Chapter 1: Introduction (The "Why")

Chapter 1 in the new format focuses strictly on establishing the project's background, purpose, boundaries, and operational definitions.

- **1.1 Project Context:** Copy the initial background and organizational context from Previous Chapter I. Ensure it clearly states the client organization (e.g., Department of Agriculture), current operational baseline, and primary "pain points".
- **1.2 Purpose and Description:** Extract the problem statement and system description paragraphs from Previous Chapter I's opening pages. Synthesize this into a concise narrative detailing what system is being proposed and how it addresses the core client challenges.
- **1.3 Objectives of the Study:** Re-number the existing objectives.
  - *1.3.1 General Objective:* The overall goal (the "what", e.g., "To design, develop, and evaluate AgriDirect...").
  - *1.3.2 Specific Objectives:* The actionable steps (the "how", listed as bullet points: a. Develop ML models, b. Design PWA farmer interface, c. Implement batch pooling, d. Conduct UAT).
- **1.4 Scope and Delimitations:** Re-organize the Previous Scope and Limitations. List included functional features, geographic focus (e.g., Province of Bulacan), and specific target user groups (Farmers, Buyers, Admins). Explicitly state non-included features under Delimitations (e.g., exclusion of direct online payment processing or logistics fulfillment).
- **1.5 Significance of the Study:** Retain the itemized list of beneficiaries (e.g., Smallholder Farmers, Institutional Buyers, Department of Agriculture, Future Researchers, NTC Academic Community).
- **1.6 Definition of Terms:** Relocate the operational definitions table/list from the end of Previous Chapter I directly into Section 1.6. Ensure terms are operationally defined within the context of the software.

Common Mistake in Chapter 1

Do **NOT** leave Literature Reviews, Technical Background, or System Architecture in Chapter 1. All literature moves to Chapter 2, and all technical specs move to Chapter 3 or 4.

2 Building Chapter 2: Review of Related Literature and Studies (The "Foundation")

Previous Chapter I contained scattered RRL and RRS sections. In the New Format, Chapter 2 becomes an independent chapter with three mandatory sections.

- **2.1 Related Literature (Foreign and Local):** Transfer literature regarding concepts, theories, policies, and frameworks (e.g., Smart Farming concepts, High-Value Crops laws, direct market theories). Separate sources into Foreign and Local sub-categories or organize them logically by theme with explicit geographic tags.
- **2.2 Related Studies (Foreign and Local):** Transfer empirical studies of existing software systems, mobile applications, and machine learning models (e.g., Project SARAI, e-commerce platforms, AI crop prediction systems).
- **2.3 Synthesis of the Reviewed Literature and Studies (NEW MANDATORY SECTION):** You must write a brand-new synthesis section. This section must NOT just repeat summaries; it must aggregate insights, identify current technology gaps, and explicitly justify why your project is necessary.

Blueprint for Writing Section 2.3 Synthesis

A strong synthesis follows a 3-part structure:

1. **Summary of Common Themes:** Identify overarching trends in existing literature (e.g., "The reviewed literature underscores a global transition toward AI-driven decision support and direct digital marketplaces...").
2. **Identification of the Research Gap:** Highlight what existing solutions lack (e.g., "However, existing local applications remain fragmented—focusing either on yield prediction or e-commerce without integrating environmental data like Project SARAI into buyer batch-pooling...").
3. **Project Positioning:** Explain how your proposed system addresses the gap (e.g., "AgriDirect fills this gap by combining machine-learning demand forecasting with location-based batch pooling specifically tailored for smallholder farmers in Bulacan.").

3 Constructing Chapter 3: Technical Background (The "Tech Stack & Architecture")

Chapter 3 establishes the technical viability of the proposed solution, moving technology descriptions and hardware/software specifications into a structured technical chapter.

- **3.1 Overview of Current Technologies:** Extract the narrative tech stack overview from Previous Chapter I. Discuss the underlying frameworks, programming languages, databases, and machine learning libraries (e.g., Python, Django, MySQL, Progressive Web Apps, Scikit-learn, Chart.js).
- **3.2 Evaluation of the Existing System (Manual or Automated) (NEW SECTION):** Formally evaluate the current baseline process utilized by the client or target demographic. Contrast manual practices (e.g., paper ledger tracking, phone-based buyer inquiries, reliance on middlemen traders) or legacy automated tools against performance, accuracy, and accessibility criteria.
- **3.3 Technical Feasibility:** Consolidate developer/user specification requirements previously scattered across the Conceptual Framework (IPO Model) and Chapter II Requirements Analysis into three explicit sub-sections:
  - *3.3.1 Hardware Specifications:* Itemize minimum and recommended hardware requirements for Development, Server/Hosting, Admin Web Access, and Mobile User Access.
  - *3.3.2 Software Specifications:* Itemize Operating Systems, Backend Frameworks, Database Engines, Frontend Libraries, ML Libraries, and Deployment Environments.
  - *3.3.3 Network/Architecture Specifications:* Transfer the **Software Architecture Diagram** and technical narrative from Previous Chapter II. Detail the 3-tier architecture (Presentation Layer, Application Layer with Django REST, Data Layer with MySQL) and network protocols (HTTPS/REST API).

4. Concrete Formatting & Writing Examples

Example: Drafting Section 3.2 (Evaluation of the Existing System)

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3.2 Evaluation of the Existing System (Manual or Automated)

Currently, agricultural trade and crop planning among smallholder farmers in Bulacan rely predominantly on a manual, intermediary-driven workflow. The existing process is evaluated below across three critical dimensions:

1. Crop Planning & Production Scheduling (Manual Baseline)
   • Existing Process: Decisions regarding planting schedules and crop selection are driven by historical intuition, neighbor practices, and tradition.
   • Identified Bottlenecks: Lack of data-driven market demand forecasting leads to periodic crop oversupply during harvest seasons, resulting in price drops and post-harvest spoilage.

2. Market Coordination & Intermediary Trading (Manual Baseline)
   • Existing Process: Farmers sell produce individually to local traders or middlemen who dictate farmgate prices and transport produce to urban hubs.
   • Identified Bottlenecks: High intermediary margins reduce farmer net earnings by 30-45%. Smallholder farmers working individually cannot meet volume thresholds required by institutional buyers (HORECA).

3. Technical Infrastructure Evaluation
   • Existing Automated Tools: Government initiatives such as DOST Project SARAI provide climate data, but data remains unintegrated with real-time commercial buyer reservation platforms.
   • System Gap: Absence of an integrated dual-platform solution that bridges environmental analytics with location-based buyer batch pooling.
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Example: Structuring Section 3.3 (Technical Feasibility Tables)

Format Section 3.3.1 and 3.3.2 cleanly using structured specification tables as shown below:

3.3.1 Hardware Specifications

Component	Minimum Specs	Recommended
Dev Processor	Intel i5 (8th Gen)	Intel i7 / Ryzen 7
Dev Memory	8 GB RAM	16 GB RAM
Dev Storage	256 GB SSD	512 GB NVMe SSD
User Mobile	Android 8.0 / iOS 13	Android 11+ / iOS 15+

3.3.2 Software Specifications

Category	Technology / Tool	Purpose
Backend	Python 3.10+, Django 4.x	Core application logic & API
Database	MySQL 8.0 / XAMPP	Relational data storage
ML Engines	Scikit-learn, Pandas	Demand & shelf-life models
Frontend	HTML5, Bootstrap 5, PWA	Responsive UI & caching

5. Where Did the Rest of Previous Chapter II Go?

A frequent point of confusion during format conversion is determining the destination of remaining Chapter II design assets (SDLC, System Diagrams, Testing, Data Dictionaries). In the New Format, these assets move directly into **Chapter 4: Methodology, Testing, and Evaluation**.

Previous Chapter II Asset	New Standard Location	Formatting Requirements
Software Development Life Cycle (Agile SDLC)	**Ch 4.1 Software Development Life Cycle**	Retain Agile diagram & sprint phase explanations (Planning, Design, Dev, Testing, Deployment, Review).

Previous Chapter II Asset	New Standard Location	Formatting Requirements
Requirements Documentation & Specs	**Ch 4.2 Requirements Analysis**	Include Functional & Non-Functional Requirements lists.
System Flowcharts & Diagrams	**Ch 4.3 System Design and Modeling** • 4.3.1 Context Diagram (Level 0) • 4.3.2 Use Case Diagram • 4.3.3 Data Flow Diagram (Level 1 & 2) • 4.3.4 Entity Relationship Diagram (ERD)	Place diagrams under explicit sub-headers 4.3.1 through 4.3.4 with complete narrative descriptions following each figure.
Data Dictionary Tables	**Ch 4.3.5 Data Dictionary**	Position data dictionaries immediately following the ERD in sub-section 4.3.5.
Pre-Testing Procedures & Requirements	**Ch 4.4 Software Testing Plan** • 4.4.1 Unit Testing • 4.4.2 Integration Testing • 4.4.3 System Testing	Expand testing descriptions into explicit Unit, Integration, and System testing sub-sections with test case matrices.
Statistical Treatment & Evaluation	**Ch 4.5 User Acceptance Testing (UAT) & Evaluation**	Incorporate sampling techniques, Likert scale interpretations, and ISO/IEC 25010 software quality evaluation metrics.

6. Master Conversion Checklist for Capstone Students

Use this actionable checklist to verify that your document conversion is 100% compliant with the New Format standard prior to proposal submission:

- ☐ **Chapter 1 Restructured:** Confirmed that Chapter 1 contains ONLY Sections 1.1 through 1.6. Removed all literature, tech stacks, and architecture diagrams from Chapter 1.
- ☐ **Objectives Numbered Correctly:** Formatted General Objective as 1.3.1 and Specific Objectives as 1.3.2. Checked that specific objectives use actionable verbs (design, develop, evaluate).
- ☐ **Definition of Terms Relocated:** Moved Definition of Terms from the end of Chapter I to Section 1.6 in Chapter 1.
- ☐ **Chapter 2 Created:** Established Chapter 2 with 2.1 (Literature), 2.2 (Studies), and 2.3 (Synthesis). Divided sources into Foreign and Local classifications.
- ☐ **Synthesis Section Written:** Authored Section 2.3 Synthesis summarizing common literature themes, identifying specific research gaps, and positioning the capstone project.
- ☐ **Chapter 3 Created:** Established Chapter 3 with 3.1 (Tech Overview), 3.2 (Existing System Evaluation), and 3.3 (Technical Feasibility).
- ☐ **Existing System Evaluated:** Authored Section 3.2 evaluating manual baseline workflows, paper processes, or legacy tools with explicit bottleneck identification.
- ☐ **Technical Feasibility Split:** Created sub-sections 3.3.1 (Hardware Specs), 3.3.2 (Software Specs), and 3.3.3 (Network/Architecture Specs with 3-tier diagram).
- ☐ **Chapter 4 Re-mapped:** Transferred SDLC (4.1), Requirements (4.2), System Diagrams & Data Dictionary (4.3), Testing Plans (4.4), and UAT/Evaluation (4.5) from Previous Chapter II to Chapter 4.
- ☐ **Formatting & Typography Verified:** Set document font to Times New Roman 12pt (or template default), updated Table of Contents page numbering, and verified APA references in Back Matters.